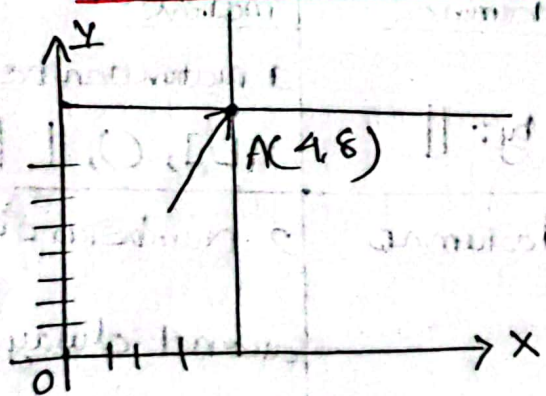


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Chapter-1 Matrix Terminology



$\vec{OA}$ can be written as $= [4, 8]$

$= \begin{pmatrix} 4 \\ 8 \end{pmatrix} \rightarrow$ matrix form

Determinant: A determinant is a function of square matrix

that reduces it to a single number. $A = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$

$$|A| = \begin{vmatrix} 4 & 1 \\ 1 & 2 \end{vmatrix} = (4 \times 2 - 1 \times 1) = 7$$

Normal vector:

$$\vec{n} = [2, 4, 1, 2]$$

$$|\vec{n}| = \sqrt{2^2 + 4^2 + 1^2 + 2^2}$$

$$= \sqrt{25} = 5$$

* Distinguish between Determinant and matrices.

<u>Determinant</u>	<u>Matrix</u>
1. Determinant, can be denoted by: $ $	1. Matrix can be denoted by: $[]$, $()$, $ $
2. Numbers of row and columns are always equal Ex: $\begin{vmatrix} 2 & 9 \\ 6 & 1 \end{vmatrix}$	2. Numbers of row and column are not always equal Ex: $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$
3. It has a definite value at. through it in an arrangement of numbers.	3. It has no definite value, it is merely an arrangement of numbers.
4. If two rows/columns are identical, a determinant vanishes.	4. Identical row (or column) may occur in a matrix.
5. Row and column can be interchanged.	5. Row and column cannot be interchanged.
6. The product of two determinants does not change the order. Ex: $\begin{vmatrix} 1 & 5 \\ 3 & 1 \end{vmatrix} \times \begin{vmatrix} 3 & 1 \\ 7 & 5 \end{vmatrix}$ $= \begin{vmatrix} 38 & 26 \\ 16 & 8 \end{vmatrix}$	6. The product of two matrices may change the order. Ex: $\begin{pmatrix} 2 & 1 \end{pmatrix}_{1 \times 2} \times \begin{pmatrix} 5 \\ 8 \end{pmatrix}_{2 \times 1} = (12)_{1 \times 1}$

Types of Matrices/

1. Square Matrix:- If $m = n$, row and column are equal

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 8 \\ 9 & 8 & 7 \end{bmatrix} \begin{matrix} \text{row} \\ \text{column} \end{matrix}$$

m, n
 $(3, 3)$

2. Rectangular:- If $m \neq n$.

$$\begin{bmatrix} 2 & 8 & 9 & 7 \\ 2 & 1 & 1 & 3 \end{bmatrix}$$

(2×4)

3. Horizontal when $(\text{row} = m < \text{column} = n)$.

$$\begin{bmatrix} 3 & 4 & 7 \\ 8 & 9 & 6 \end{bmatrix}$$

(2×3)

4. Vertical:- when $(\text{row} = m > \text{column} = n)$.

$$\begin{bmatrix} 1 & 1 \\ 3 & 7 \\ 8 & 9 \end{bmatrix}$$

3×2

5. Row:- Only row

$$[1 \ 2 \ 3 \ 1]$$

6. column $\begin{bmatrix} 3 \\ 1 \\ 7 \end{bmatrix}$
 Null $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

7. Unit matrix: In the leading diagonal elements are non-zero but not more than 1.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

8. Diagonal: A square matrix, all elements are zero except the main or leading diagonal

$$B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

Trace :-

The sum of the diagonal elements of a square matrix is called the trace of the matrix. $B = 2 + 7 + 9 = 18$

Sub-matrix :

$$\begin{bmatrix} 1 & 7 & 10 \\ 8 & 9 & 9 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 7 \\ 8 & 9 \end{bmatrix}$$

Periodic matrix:- $A^{k+1} = A$.

Idempotent matrix:- $A^2 = A$

Nilpotent matrix:- $A^m = 0$; $A^{m-1} \neq 0$.

Involutory matrix:- $A^2 = I$

Complex conjugated matrix:- $A = \begin{bmatrix} 1+i & 2+3i \\ 2 & 3i \end{bmatrix}$

$$\bar{A} = \begin{bmatrix} 1-i & 2-3i \\ 2 & -3i \end{bmatrix}$$

Real matrix:- $A = \bar{A}$

Imaginary:- $A = -\bar{A}$.

Transpose of matrix: $A = \begin{bmatrix} 2 & 7 & 8 \\ 8 & 9 & 10 \end{bmatrix}$ $A^T = \begin{bmatrix} 2 & 8 \\ 7 & 9 \\ 8 & 10 \end{bmatrix}$

Symmetric: $A = (a_{ij})$; $a_{ij} = a_{ji}$ $[A^T = A, \text{condition}]$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Here

$$\begin{aligned} a_{11} &= a_{11} = 1 \\ a_{21} &= a_{12} = 2 \\ a_{31} &= a_{13} = 3 \end{aligned}$$

$$\begin{aligned} a_{23} &= a_{32} = 5 \\ a_{33} &= a_{33} = 6 \\ a_{22} &= a_{22} = 4 \end{aligned}$$

Skew Symmetric matrix:- A square matrix is called skew

symmetric matrix $a_{ij} = -a_{ji}$.

$$A = \begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{bmatrix}$$

$$a_{ij} + a_{ji} = 0$$

$$\text{Ex: } a_{22} + a_{22} = 0, [0 + 0 = 0]$$

Selfadjoint: $A^H = \overline{A^T} = A$. = Hermitian matrix

Skew-Hermitian matrix:-

$$A^* = -A$$

$$a_{ij} = -\overline{a_{ji}}$$

$$\begin{vmatrix} i & 2ti \\ -2ti & 3i \end{vmatrix}$$

Unitary matrix:-

$$A^{-1} = A^{-H}$$

Singular matrix: $|A| = 0$.

Q-1 Show that $A = \begin{bmatrix} 2 & -2 & -1 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is idempotent

Soln: For idempotent matrix,

condition, $A^2 = A$

$$\therefore A^2 = A \cdot A$$

$$= \begin{bmatrix} 2 & -2 & -1 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 & -2 & -1 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 4-2-1 & -4+6+3 & -2-8+3 \\ -2-3+4 & 2+9-8 & 4+12-12 \\ 2+2-3 & -2-6+6 & -4-8+9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

$$= A$$

So, Matrix A is idempotent.

Q-2 $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is involutory matrix, prove it.

for involutory matrix,

$$A^2 = I$$

L.H.S

$$A^2 = A \cdot A$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I = \text{R.H.S}$$

(proved)

Q-3

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

is a singular matrix or not.

Sol:-

$$|A| = 1(24-25) - 2(12-15) + 3(10-12)$$

$$= -1 + 6 - 6$$

$$= -1$$

As $|A| \neq 0$, hence the matrix is not singular

Ex-4 $A = \begin{pmatrix} 3 & 2+i \\ 2-i & 1 \end{pmatrix}$ is a Hermitian matrix.

for Hermitian matrix;

$$\therefore A^H = \overline{A^T} = A.$$

given matrix;

$$A = \begin{pmatrix} 3 & 2+i \\ 2-i & 1 \end{pmatrix}$$

$$\text{or, } A^T = \begin{pmatrix} 3 & 2-i \\ 2+i & 1 \end{pmatrix}$$

$$\text{So, } \overline{A^T} = \begin{pmatrix} 3 & 2+i \\ 2-i & 1 \end{pmatrix} \\ = A$$

So, it's a Hermitian matrix. (proved)

Ex-8 If $A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$ Show that $A^n = \begin{pmatrix} 1+2n & -n \\ n & 1-2n \end{pmatrix}$

Also, find A^{11}

Soln:

$$A^2 = A \cdot A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \\ = \begin{pmatrix} 9-4 & -12+4 \\ 3-1 & -4+1 \end{pmatrix}$$

$$A^2 = \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1+2 \cdot 2 & -4 \cdot 2 \\ 2 & 1-2 \cdot 2 \end{bmatrix}$$

$$\therefore A^3 = \begin{bmatrix} 1+2 \cdot 3 & -4 \cdot 3 \\ 3 & 1-2 \cdot 3 \end{bmatrix}$$

... ..
... ..

$$A^n = \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$$

$$A^{n+1} = \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix} \cdot \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 3(1+2n) - 4n & -4(1+2n) + 4n \\ 3n + 1 - 2n & -4n - 1 + 2n \end{bmatrix}$$

$$= \begin{bmatrix} 3+6n-4n & -4-8n+4n \\ n+1 & -4n-1+2n \end{bmatrix}$$

$$\therefore A^{n+1} = \begin{bmatrix} 3+2n & -4-4n \\ n+1 & -2n-1 \end{bmatrix}$$

$$A^{n+1} = \begin{bmatrix} 1+2n+2 & -4(n+1) \\ n+1 & 1-2n-2 \end{bmatrix}$$

Inductive Hypothesis

$$= \begin{bmatrix} 1+2(n+1) & -4(n+1) \\ n+1 & 1-2(n+1) \end{bmatrix}$$

Ans

Ex-9

$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ Show $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$, also find A^{n+1}

Soln:

$$A^2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1+0 & 1+1 \\ 0 & 1+1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

$$A^2 = A \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

$$A^3 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$$

$$\dots \dots \dots$$

$$\therefore A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

$$\therefore A^{n+1} = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1+n \\ 0 & 1 \end{pmatrix}$$

$$\therefore A^{n+1} = \begin{bmatrix} 1 & n+1 \\ 0 & 1 \end{bmatrix} \underline{A}$$

Inverse matrix

The inverse of a matrix, $A^{-1} = \frac{\text{Adjoint of } A}{|A|}$

$|A| \rightarrow$ determinant of matrix A

Ex-17

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \quad \text{where } |A| \neq 0$$

$$|A| = 1(24 - 25) - 2(12 - 15) + 3(10 - 12)$$

$$= -1 + 6 - 6$$

$$= -1 < 0$$

Now,

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 4 & 5 \\ 5 & 6 \end{vmatrix} = -1 \quad A_{13} = (-1)^{1+3} \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} = -2$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix} = 3$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = 3$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 3 \\ 3 & 6 \end{vmatrix} = -3$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 3 & 5 \end{vmatrix} = 1$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 2 & 4 \\ 4 & 5 \end{vmatrix} = -2$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix} = 1$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 0$$

$$A^c = \begin{bmatrix} -1 & 3 & -2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{bmatrix}$$

Adjoint of $A = (A^c)^t$

$$\begin{bmatrix} -1 & 3 & -2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{bmatrix}$$

$$A^{-1} = \frac{(A^c)^t}{|A|} = \frac{1}{-1} \begin{bmatrix} -1 & 3 & -2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{bmatrix}$$

Perfection test

$$AA^{-1} = I$$

$$= \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1-6+6 & -3+6-3 & 2-2+0 \\ 2-12+10 & -6+12-5 & 4-4+0 \\ 3-15+12 & -9+15-6 & 6-5+0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Ex-18

$$A = \begin{pmatrix} a+ib & c+id \\ -c+id & a-ib \end{pmatrix} \quad \text{find } A^{-1}.$$

$(a^2+b^2+c^2+d^2=1)$

$$\begin{aligned} |A| &= (a+ib)(a-ib) + \{c^2 - i^2 d^2\} \\ &= (a^2 - i^2 b^2 + c^2 - i^2 d^2) \quad [\because -i^2 = 1] \\ &= a^2 + b^2 + c^2 + d^2 = 1 > 0 \end{aligned}$$

$$A_{11} = (-1)^{1+1} (a-ib) = a-ib$$

$$A_{12} = (-1)^{1+2} (-c+id) = c-id$$

$$A_{21} = (-1)^{2+1} (c+id) = -c-id$$

$$A_{22} = (-1)^{2+2} (a+ib) = a+ib$$

$$A^c = \begin{pmatrix} a-ib & c-id \\ -c-id & a+ib \end{pmatrix}$$

$$(A^c)^t = \begin{pmatrix} a-ib & -c-id \\ c-id & a+ib \end{pmatrix}$$

$$A^{-1} = \frac{(A^c)^t}{|A|} = \frac{(A^c)^t}{1}$$

$$= \begin{pmatrix} a-ib & -c-id \\ c-id & a+ib \end{pmatrix}$$

Perfection test:

$$AA^{-1} = \begin{pmatrix} a+ib & c+id \\ -c+id & a-ib \end{pmatrix} \begin{pmatrix} a-ib & -c-id \\ c-id & a+ib \end{pmatrix}$$

$$= \begin{pmatrix} a^2 - i^2 b^2 + c^2 - i^2 d^2 & (a+ib)(-c-id) + (c+id)(a+ib) \\ (-c+id)(a-ib) + (a-ib)(c-id) & - (i^2 d^2 - c^2) + a^2 - i^2 b^2 \end{pmatrix}$$

$$= \begin{pmatrix} a^2 + b^2 + c^2 + d^2 & - (a+ib)(c+id) + (a+ib)(c+id) \\ - (a-ib)(c-id) + (a-ib)(c-id) & a^2 + b^2 + c^2 + d^2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Characteristic roots or latent roots

Any root are called characteristic root or latent roots or

Eigen values or characteristic values

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 3 & \lambda-2 \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(\lambda-2) - 6 = 0$$

$$\lambda^2 - 3\lambda - 4 = 0$$

$\lambda = 4, -1$; There are characteristic

root or latent root or Eigen values of A

Ex-19: Find the characteristic roots or latent roots or Eigen values

of matrix $A = \begin{pmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{pmatrix}$

$$A - \lambda I = \begin{pmatrix} \cos \theta - \lambda & -\sin \theta \\ -\sin \theta & -\cos \theta - \lambda \end{pmatrix} = \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos \theta - \lambda & -\sin \theta \\ -\sin \theta & -\cos \theta - \lambda \end{pmatrix} = \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos \theta - \lambda & -\sin \theta \\ -\sin \theta & -\cos \theta - \lambda \end{pmatrix}$$

$$\therefore |A - \lambda I| = (\cos \theta - \lambda)(-\cos \theta - \lambda) - \sin^2 \theta$$

$$= 1(\cos^2 \theta - \lambda^2) - \sin^2 \theta$$

$$= +\lambda^2 - 1$$

$$\therefore |\lambda - \lambda I| = 0$$

$$\text{or } \lambda^2 - 1 = 0$$

$$\text{or } \lambda^2 = 1$$

$$\lambda = \pm 1$$

Linear Equation

$$\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ \uparrow & \uparrow & \uparrow \\ 2x + 3y + z = 7 \\ a_{21} & a_{22} & a_{23} \\ 4x + 5y + 7z = 9 \\ a_{31} & a_{32} & a_{33} \\ x + 9y + z = 0 \end{array}$$

} Non-homogeneous system of linear equations.

Here, a_{11} has non-zero coefficient. So that, $a_{11} \neq 0$.

$$L_1 = -a_{12}L_1 + a_{11}L_1$$

$$L_2 = -a_{21}L_1 + a_{11}L_2$$

$$L_3 = -a_{31}L_1 + a_{11}L_3$$

Ex-24 Solve the system;

$$2x + y - z = 10 \dots\dots\dots L_1$$

$$3x + 2y + z = 1 \dots\dots\dots L_2$$

$$5x + 4y + 3z = 4 \dots\dots\dots L_3$$

Here, $a_{11} = 2$, $a_{12} = 1$, $a_{13} = -2$,

$a_{21} = 3$, $a_{22} = 2$, $a_{23} = 1$

$a_{31} = 5$, $a_{32} = 4$, $a_{33} = 3$

1st time,

$i = 2$, $L_2 = -a_{21}L_1 + a_{11}L_2$

$$L_2$$

$$= -3(2x - y - 2z - 10) + 2(3x + 2y + 2z - 1)$$

$$= -6x - 3y + 6z + 30 + 6x + 4y + 4z - 2$$

$$= -3y + 4y + 10z + 28 = y + 10z + 28$$

L_3

$$= -5(2x - y - 2z - 10) + 2(5x + 4y + 3z - 4)$$

$$= -10x - 5y + 10z + 50 + 10x + 8y + 6z - 8$$

$$= 3y + 16z + 42$$

Thus we obtain the following new system

$$2x - y - 2z = 10$$

$$y + 10z = -28$$

$$3y + 16z = -42$$

Here, we have

$$a_{11} = 2, a_{21} = 10, a_{31} = 3, a_{22} = 16.$$

2nd time

$$E_2, L_2 = -3(y + 10z + 28) + 1(3y + 16z + 42)$$

$$= -3y - 30z - 84 + 3y + 16z + 42$$

$$= -14z - 42$$

Thus we obtain the following system,

$$2x + y - 2z = 10$$

$$y + 10z = -28$$

$$-14z = 42$$

Now, $-14z = 42$

$$\therefore z = -3$$

Again, $y + 10z = -28$

$$y + 10(-3) = -28$$

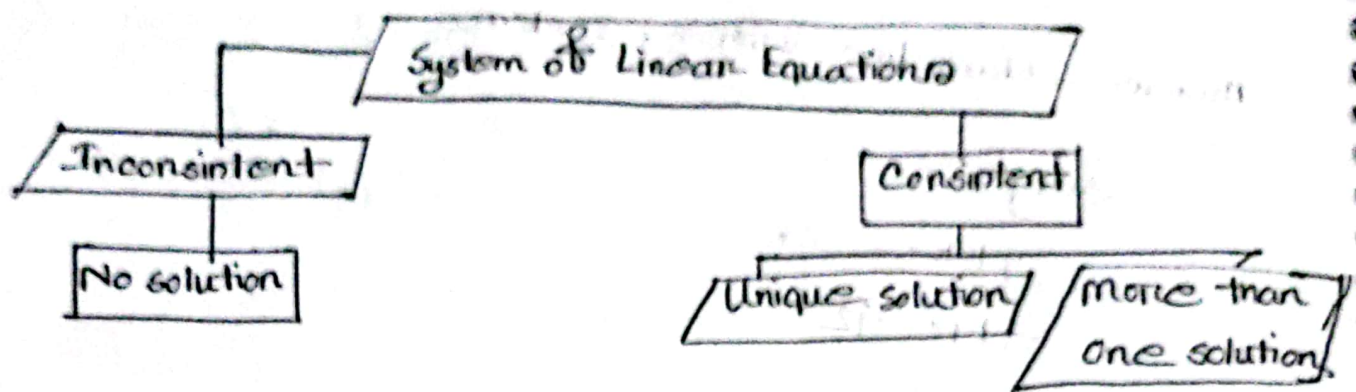
$$\therefore y = -2$$

Again, $2x + y - 2z = 10$

$$\therefore 2x - y - 2(-3) = 10$$

$$(2x - y - 2(-3)) \Rightarrow x = 1$$

$$\therefore (x, y, z) = (1, -2, -3)$$



Ex-24 Solve the system,

$$2x + y - 2z = 10 \quad \dots L_1$$

$$3x + 2y + 2z = 1 \quad \dots L_2$$

$$5x + 4y + 3z = 4 \quad \dots L_3$$

Here,

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

1st time,

$$i=2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -3(2x + y - 2z - 10) + 2(3x + 2y + 2z - 1)$$

$$= -6x - 3y + 6z + 30 + 6x + 4y + 4z - 2$$

$$= y + 10z + 28$$

$$i=3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$= -5(2x + y - 2z - 10) + 2(5x + 4y + 3z - 4)$$

$$= -10x - 5y + 10z + 50 + 10x + 8y + 6z - 8$$

$$= 3y + 16z + 42$$

Thus we obtain the following new system

$$2x + y - 2z = 10$$

$$y + 10z = -28 \quad \text{--- L}_1$$

$$3y + 16z = -42 \quad \text{--- L}_2$$

2nd time :

$$2x + y - 2z = 10$$

we have,

$$L_1 \rightarrow a_{11}L_1 + a_{21}L_1$$

$$i=2, L_2 \rightarrow a_{21}L_1 + a_{11}L_2$$

$$= -3(y + 10z + 28) + 1(3y + 16z + 12)$$

$$= -11z - 42$$

Now,

$$-11z = 42, \text{ or } z = -3$$

$$\text{Again, } y + (10 \times -3) = -28$$

$$\therefore y = 2$$

$$\text{Again, } 2x + y - 2z = 10$$

$$\therefore x = 1$$

$$\text{Ans: } (x, y, z) = (1, 2, -3)$$

Ex-26

Solve the system

$$2x + 3y - 2z = 5 \quad \text{---} \quad L_1$$

$$x - 2y + 3z = 2 \quad \text{---} \quad L_2$$

$$4x - y + 4z = 1 \quad \text{---} \quad L_3$$

Soln:

1st time,

$$L_i \rightarrow a_{i1}L_1 + a_{i2}L_2$$

$$i=2, L_2 \rightarrow -a_{21}L_1 + a_{22}L_2$$

$$= -1(2x + 3y - 2z) + 2(1x - 2y + 3z - 2)$$

$$= -2x - 3y + 2z + 5 + 2x - 4y + 6z - 4$$

$$= -7y + 8z + 1$$

$$i=3, L_3 \rightarrow -a_{31}L_1 + a_{32}L_2$$

$$= -4(2x + 3y - 2z) + 2(1x - y + 3z - 2)$$

$$= -8x - 12y + 8z + 20 + 2x - 2y + 6z - 4$$

$$= -6x - 14y + 14z + 16$$

Thus we obtain the following system

$$2x + 3y - 2z = 5$$

$$-7y + 8z = -1 \quad \text{--- } L_1$$

$$-14y + 16 = -18 \quad \text{--- } L_2$$

2nd time

$$i=2, L_2 \rightarrow a_{21}L_1 + a_{11}L_2$$

$$= (-14) \left(\overset{-7y+8z+1}{2x+3y-2z=5} \right) + 7(-7y+8z+1)$$

$$= \cancel{-28x - 42y + 28z + 70} - 14y + 16z + 2$$

$$= \cancel{0} - 98y + 112z + 14 + 98y - 112z - 126$$

$$= -112$$

(1) Thus we obtain the following system

$$2x + 3y - 2z = 5$$

$$-7y + 8z = -1$$

$$0 = 112$$

$$\text{Here, } 0x + 0y + 0z = 112$$

$$\Rightarrow 0 = 112$$

but $112 \neq 0$ It is impossible (inconsistent)

So, the system is inconsistent and has no solution.

Ex-32 find the condition on λ and u so that the following system of linear equations will have i) a unique solution (ii) more than one solution (iii) no solution.

$$2x + 3y + z = 5 \quad \text{--- } L_1$$

$$3x - y + \lambda z = 2 \quad \text{--- } L_2$$

$$x + 7y - 6z = u \quad \text{--- } L_3$$

Soln:

we have,

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$i=2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\rightarrow -3(2x + 3y + z - 5) + 2(3x - y + \lambda z - 2)$$

$$\rightarrow -6x - 9y - 3z + 15 + 6x - 2y + 2\lambda z - 4$$

$$\rightarrow -11y - z(3 - 2\lambda) + 11 \quad \text{--- (i)}$$

$$i=3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$\rightarrow -1(2x + 3y + z - 5) + 2(x + 7y - 6z - u)$$

$$\rightarrow -2x - 3y - z + 5 + 2x + 14y - 12z - 2u$$

$$\rightarrow 11y - 13z + 5 - 2u$$

Thus we obtain the following system,

$$\begin{aligned} 2x + 3y + z &= 5 \\ -11y + (2\lambda - 3)z &= -11 \quad \text{--- } L_1 \\ \cancel{-22\lambda + 11y - 13z} &= -5 + 2u \quad \text{--- } L_2 \end{aligned}$$

2nd time:-

$$i=2 \rightarrow a_{21}L_1 + a_{22}L_2$$

$$L_2 \rightarrow -11 \{ -11y + (2\lambda - 3)z + 11 \} + (-12)(-11y - 13z + 5 + 2u)$$

$$L_2 \rightarrow 121y - 22\lambda + 33z - 121 - 121y + 143z - 55 + 22u$$

$$L_2 \rightarrow -22\lambda + 176z - 176 + 22u$$

$$\cancel{(-\lambda + 8)z}$$

Thus we obtain the following system

$$\begin{aligned} 2x + 3y + z &= 5 \\ -11y + (2\lambda - 3)z &= -11 \\ -22\lambda + 16z &= 176 - 22u \end{aligned}$$

$$\text{or, } -(\lambda - 8)z = 8 - u$$

$$\text{or, } (8 - \lambda)z = 8 - u.$$

The above system is in echelon form. Now we consider the following three cases.

a) for unique solution, the coefficient of z in the 3rd equation must be non-zero, that is, $\lambda \neq 8$ and u have any value. Therefore, the given system has unique solution for $\lambda \neq 8$ and arbitrary values of u .

b) If $\lambda = 8$, $u = 8$, then third equation of the above system vanishes, and the system will be in echelon form having two equations in three unknowns, then the system has $(3-2) = 1$ free variable, which is z . Hence the given system has more than one solution for $\lambda = 8$ and $u = 8$.

c) If $\lambda = 8$ and $u \neq 8$, then the third equation of the above system is of the form $0 = u - 8$, which is inconsistent. Thus the system has no solution for $\lambda = 8$ and $u \neq 8$.

Linear independence & dependence

The vectors $\{v_1, v_2, v_3, \dots, v_n\}$ are only when the scalars $a_i = (a_1, a_2, a_3, \dots, a_n)$ are all equal to zero. The vectors are linearly dependent if the equation has a solution when at least one of the scalars is not zero.

Ex-33 Determine whether the following vectors in $\mathbb{R}^2$ are linearly dependent or independent.

$$v_1 = \begin{bmatrix} -1 \\ 3 \end{bmatrix} \quad v_2 = \begin{bmatrix} 5 \\ 6 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Soln: Suppose we have a linear combination of the vectors equal to 0.

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = 0$$

$$a_1 \begin{bmatrix} -1 \\ 3 \end{bmatrix} + a_2 \begin{bmatrix} 5 \\ 6 \end{bmatrix} + a_3 \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -a_1 \\ 3a_1 \end{bmatrix} + \begin{bmatrix} 5a_2 \\ 6a_2 \end{bmatrix} + \begin{bmatrix} a_3 \\ 4a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -a_1 + 5a_2 + a_3 \\ 3a_1 + 6a_2 + 4a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow -a_1 + 5a_2 + a_3 = 0 \quad \text{--- } L_1$$

$$3a_1 + 6a_2 + 4a_3 = 0 \quad \text{--- } L_2$$

To reduce it to a simpler system for each $i > 1$, apply the operation:

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\rightarrow -3(-a_1 + 5a_2 + a_3) + 4(3a_1 + 6a_2 + 4a_3)$$

$$\rightarrow -15a_2 - 3a_3 + 3a_1 - 3a_1 - 6a_2 - 4a_3$$

$$\rightarrow -21a_2 - 7a_3$$

Thus we obtain the following new system

$$-a_1 + 5a_2 + a_3 = 0$$

$$-21a_2 - 7a_3 = 0$$

In echelon form, there are only two equations for three unknowns, then the system has a non-zero solution and in particular $(3-2) = 1$ free variable, which is a_3 .

Since, a_3 is a free variable, we can choose it to be anything.

In particular, we can choose it to be non-zero. Therefore, the vectors are linearly dependent.

Ex-34 Determine whether the following vectors $\mathbb{R}^3$ are

linearly dependent or independent.

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad v_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Soln:

Suppose we have a linear combination of the vectors equal to 0.

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = 0$$

$$\Rightarrow \begin{bmatrix} a_1 \\ 2a_1 \\ 3a_1 \end{bmatrix} + \begin{bmatrix} -2a_2 \\ a_2 \\ 0 \end{bmatrix} + \begin{bmatrix} a_3 \\ 0 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_1 - 2a_2 + a_3 \\ 2a_1 + a_2 + 0 \cdot a_3 \\ 3a_1 + 0 \cdot a_2 + a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$a_1 - 2a_2 + a_3 = 0$$

$$2a_1 + a_2 = 0$$

$$3a_1 + a_3 = 0$$

To reduce it to a simpler system for each $i \geq 1$ applies

the operation:

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$L_2 \rightarrow -a_{21}L_1 + a_{12}L_2$$

$$\rightarrow -2(a_1 - 2a_2 + a_3) + 1(2a_1 + a_2)$$

$$\rightarrow -2a_1 - 4a_2 - 2a_3 + a_2 + 2a_1$$

$$\rightarrow -3a_2 - 2a_3$$

$$i=3,$$

$$L_3 \rightarrow -a_3 L_1 + a_1 L_2$$

$$\rightarrow -3(a_1 - 2a_2 + a_3) + 1(3a_1 + a_3)$$

$$\rightarrow -3a_1 + 6a_2 - 3a_3 + 3a_1 + a_3$$

$$\rightarrow 6a_2 - 2a_3$$

Thus we obtain the following new system

$$a_1 - 2a_2 + a_3 = 0$$

$$5a_2 - 2a_3 = 0 \quad \text{--- } L_1$$

$$6a_2 - 2a_3 = 0 \quad \text{--- } L_2$$

To reduce it to a simpler system for each $i \geq 1$ applied the operation:

$$i=2, L_2 \rightarrow -a_2 L_1 + a_1 L_2$$

$$\rightarrow -6(5a_2 - 2a_3) + (5)(6a_2 - 2a_3)$$

$$\rightarrow -30a_2 + 12a_3 + 30a_2 - 10a_3$$

$$\rightarrow 2a_3$$

Thus we obtain following new system,

$$a_1 - 2a_2 + a_3 = 0$$

$$5a_2 - 2a_3 = 0$$

$$2a_3 = 0$$

$$\therefore a_3 = 0$$

$$a_2 = 0$$

$$a_1 = 0$$

Therefore, vectors are linearly independent.

Ex-35 Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $f(t) = 2t^2 - 3t + 7$, $g(t) = t^2 - 5t - 2$

$f(A)$, $g(A)$.

$$\therefore f(A) = 2A^2 - 3A + 7I$$

$$g(A) = A^2 - 5A - 2I$$

$$f(A) = 2 \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 3 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 18 & 44 \\ 21 & 39 \end{bmatrix}$$

$$g(A) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 5 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Eigen vector / Value

An Eigen vector is a non zero vector which satisfies $A\vec{v} = \lambda\vec{v}$

A = square matrix, $\vec{v}$ = eigenvector, λ = eigen value.

E36 Determine which of the following vectors are Eigen

vectors for $A = \begin{pmatrix} 1 & 2 \\ -1 & 7 \end{pmatrix}$. Give graphical analysis.

a $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ b $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ c $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ d $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$ e $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ f $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Ans:

We have the relation between Eigen vector & Eigen values of a matrix, $A\vec{x} = \lambda\vec{x}$

where A is given matrix, λ is the eigen value of the given matrix, x is the corresponding Eigenvectors of the matrix.

Q. tent:

Given $A = \begin{pmatrix} 1 & 2 \\ -1 & 7 \end{pmatrix}$ $x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$\therefore Ax = \begin{pmatrix} 1 & 2 \\ -1 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1-2 \\ -1+7 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$A\vec{x} = \lambda\vec{x}$ not satisfied since $\vec{x}$ is not returned

Test b: $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 2+6 \\ -8+21 \end{pmatrix} = \begin{pmatrix} 8 \\ 13 \end{pmatrix}$$

$AX = \lambda X$ is not satisfied, since λ is not returned.

Test c: $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1+4 \\ -4+14 \end{pmatrix} = \begin{pmatrix} 5 \\ 10 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

~~$AX = \lambda X$~~ $\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 A X λ X

$\therefore AX = \lambda X$ satisfied, since λ is returned

Test d: $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$

$$AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2+4 \\ -8+14 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 A X λ X

$\therefore AX = \lambda X$ satisfied since λ is returned

Test f:

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+2 \\ -4+7 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $A \quad X \quad \lambda \quad X$

$\therefore AX = \lambda X$ is satisfied since λ is returned.

Test e:

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$AX = \begin{pmatrix} 2+8 \\ -8+28 \end{pmatrix} = \begin{pmatrix} 10 \\ 20 \end{pmatrix} = 5 \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 5 \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

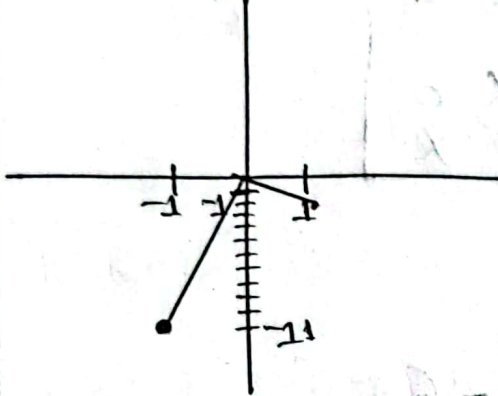
$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $A \quad X \quad \lambda \quad X$

$AX = \lambda X$ is satisfied, since λ is returned.

graphical analysis:

case (A) $A = \begin{pmatrix} 1 & 2 \\ -1 & 7 \end{pmatrix}$ $X = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

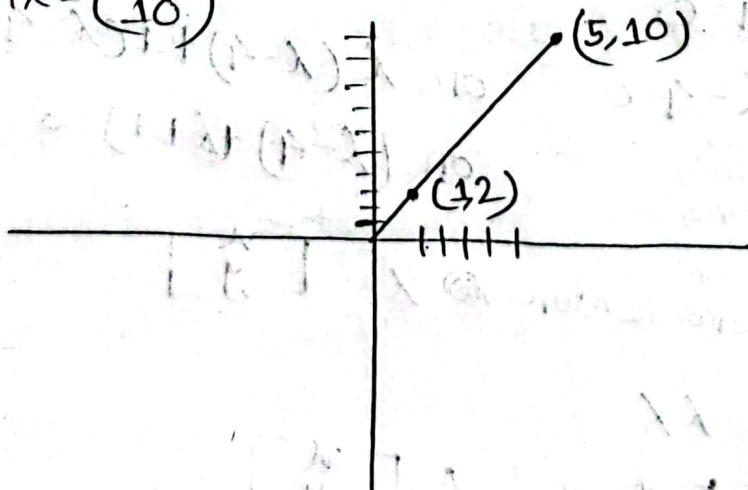
$AX = \begin{pmatrix} -1 \\ -11 \end{pmatrix}$



The product vector AX is not same direction as X . So, X is not Eigenvector of A .

case (C) $A = \begin{pmatrix} 1 & 2 \\ -1 & 7 \end{pmatrix}$ $X = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$AX = \begin{pmatrix} 5 \\ 10 \end{pmatrix}$



The product vector AX is in the same direction as X . So, X is an Eigen vector of A .

Ex-37 Find the Eigen values and Eigen vectors of matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$$

Soln: $A - \lambda I = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{bmatrix}$$

$$\therefore |A - \lambda I| = (1-\lambda)(2-\lambda) - 6$$
$$= 2 - \lambda - 2\lambda + \lambda^2 - 6$$

$$= \lambda^2 - 3\lambda - 4$$

$$\therefore |A - \lambda I| = 0$$

$$\text{or } \lambda^2 - 3\lambda - 4 = 0$$

$$\text{or } \lambda^2 - 4\lambda + \lambda - 4 = 0$$

$$\text{or } \lambda(\lambda-4) + 1(\lambda-4) = 0$$

$$\text{or } (\lambda-4)(\lambda+1) = 0 \therefore \lambda = 4, -1$$

Let, A non-zero vector $x = \begin{bmatrix} x \\ y \end{bmatrix}$

we have $Ax = \lambda x$

$$\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} x+2y \\ 3x+2y \end{bmatrix} = -1 \begin{bmatrix} x \\ y \end{bmatrix} \quad [\text{when } \lambda = -1]$$

$$\begin{bmatrix} x+2y \\ 3x+2y \end{bmatrix} = \begin{bmatrix} -x \\ -y \end{bmatrix}$$

Thus we obtain the system of equations

$$\begin{aligned} x+2y &= -x \\ 3x+2y &= -y \end{aligned} \Rightarrow \begin{aligned} 2x+2y &= 0 \\ 3x+3y &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} x+y &= 0 & \text{--- } L_1 \\ x+y &= 0 & \text{--- } L_2 \end{aligned}$$

$$L_1 \rightarrow a_{11}L_1 + a_{11}L_1$$

$$i=2, L_2 \rightarrow -a_{21}L_1 + a_{22}L_2$$

$$= -(x+y) + (x+y)$$

$$L_2 \rightarrow 0$$

Thus we obtain the following system

$$x+y=0$$

$$0=0$$

2nd equation can be deleted without affecting the solution.

So we get, $(x+y=0)$

In echelon form, there are only one equation in two unknowns

then the system has non-zero solution and in particular

$(2-1)=1$ free variable, which is y .

$$\text{Let } y=2, x=-2$$

$x = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ is a non-zero Eigenvector to
belonging to the eigen value $\lambda = -1$

Again for $\lambda = 4$,

$$AX = \lambda X$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 4 \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} x+2y \\ 3x+2y \end{bmatrix} = \begin{bmatrix} 4x \\ 4y \end{bmatrix}$$

Then we obtain the system of equations

$$x+2y = 4x$$

$$3x+2y = 4y$$

$$\Rightarrow -3x + 2y = 0$$

$$3x - 2y = 0$$

$$\Rightarrow \begin{matrix} 3x - 2y = 0 \\ 3x - 2y = 0 \end{matrix}$$

Since both equations are identical, we can disregard one of them. So, $3x - 2y = 0$

In echelon form, there are only one equation in two unknowns, then the system has non-zero solution in particular $(2-1) = 1$ free variable, which is y .

$$\text{Let, } y = 3$$

$$3x = 6 \Rightarrow x = 2$$

$x = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ is a non-zero eigenvector belonging to the eigenvalue $\lambda = 4$.

$\therefore -1, 4$ are the eigen values of matrix A and their respective eigen vectors are $\begin{bmatrix} -2 \\ 2 \end{bmatrix}$ & $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

Let P be the matrix whose columns are the eigen vectors ~~of~~ of the given matrix $\therefore P = \begin{bmatrix} -2 & 2 \\ 2 & 3 \end{bmatrix}$

Cayley-Hamilton theorem

Statement: Every square matrix satisfies its characteristic equation or if $|A - \lambda I| = (-1)^n [\lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots$

$+ a_n]$ be the characteristic polynomial of $n \times n$ matrix

$A = [a_{ij}]$, then the matrix equation

$$x^n + a_1 x^{n-1} + \dots - a_n I = 0$$

Ex-47 $A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$ A^{-1} by Cayley-Hamilton.

Soln:

Hence we have,

$$A - \lambda I = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}$$

$$|A - \lambda I| = (1-\lambda)(3-\lambda) - 8 = 3 - 3\lambda - \lambda + \lambda^2 - 8 = \lambda^2 - 4\lambda - 5$$

$\therefore$ Characteristic equation of A is $|A - \lambda I| = 0$

$$\text{or } \lambda^2 - 4\lambda - 5 = 0 \quad \text{--- (i)}$$

Replace λ by A in eqn (i).

$$A^2 - 4A - 5I$$

$$= \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4 \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+8 & 4+12 \\ 2+8 & 3+9 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 9-4-5 & 16-16-0 \\ 8-8-0 & 12-12-5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{That is } A^2 - 4A - 5I = 0 \quad \text{--- (ii)}$$

Hence the given matrix A satisfies its characteristic equation (i) i.e. satisfies Cayley-Hamilton Theorem.

Again, from (ii)

$$A^2 - 4A - 5I = 0 \quad \text{given}$$

$$A^2 - 4A = 5I \quad \text{--- (iii)}$$

Multiplying $A^{-1} \times$ (iii) both sides

$$A \cancel{A} - 4 \cancel{A} = 5IA^{-1} \quad A^2 A^{-1} - 4AA^{-1} = 5IA^{-1}$$

$$\therefore \cancel{A} - \frac{1}{5} \quad \text{or } A - 4I = 5A^{-1} \quad \left[\because AA^{-1} = I \right]$$

$$A - 4I = 5A^{-1}$$

$$\text{or } A^{-1} = \frac{1}{5} (A - 4I) \quad \text{--- (N)}$$

$$\therefore A^{-1} = \frac{1}{5} \left\{ \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

$$= \frac{1}{5} \left\{ \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \right\}$$

$$= \frac{1}{5} \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

Verification:

$$AA^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} -3+8 & 4-4 \\ -6+6 & 8-3 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

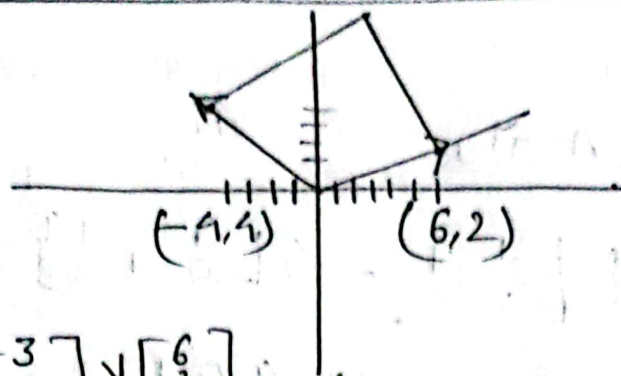
$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Ex-49 find the area of parallelogram defined by the column

vectors $u = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$ & $v = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

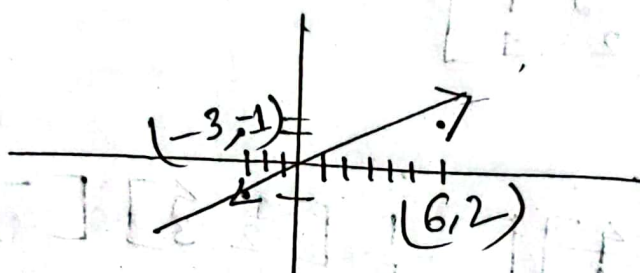
Ans: $\left| \text{determinant} \begin{bmatrix} -4 & 6 \\ 1 & 2 \end{bmatrix} \right| = |-32|$

$$= 32 \text{ square units}$$



Q-50 $u = \begin{bmatrix} -3 \\ -1 \end{bmatrix} \vee \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

$$|\det \begin{bmatrix} -3 & 6 \\ -1 & 2 \end{bmatrix}| = |0| = 0 \text{ square unit}$$



Matrix Decomposition Theorem

Eigen / Diagonal $A = PDP^{-1}$

Q-57 Examine the eigen decomposition for the matrix

$$A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$$

$$\text{Let } A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$$

Soln:

$$\therefore (A - \lambda I) = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{pmatrix}$$

$$= (4-\lambda)(4-\lambda) - 1 = 16 + \lambda^2 - 8\lambda - 1 \\ = \lambda^2 - 8\lambda + 15$$

$$|A - \lambda I| = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 15 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda - 3\lambda + 15 = 0$$

$$\Rightarrow \lambda(\lambda - 5) - 3(\lambda - 5) = 0$$

$$\therefore \lambda = 5, 3$$

$$\therefore D = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$$

Now find out eigen vector of matrix A

Let a non zero vector $v = \begin{bmatrix} x \\ y \end{bmatrix}$

We know,

$$Av = \lambda v$$

$$\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 3 \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 4x + y \\ x + 4y \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \end{pmatrix}$$

$$\Rightarrow 4x + y = 3x$$

$$x + 4y = 3y$$

$$\Rightarrow \begin{matrix} x + y = 0 \\ x + y = 0 \end{matrix}$$

$$\Rightarrow \boxed{x} + y = 0$$

In echelon form, there are only one equation for two unknowns, then system has a non-zero solution and in particular $(2-1)=1$ free variable, which is

$$\text{Let } y=1 \Rightarrow x=-1$$

$$\therefore \text{Eigenvector } v_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

for $\lambda=5$

we have

$$Av = \lambda v$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 5 \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = \begin{pmatrix} 5x \\ 5y \end{pmatrix}$$

$$\Rightarrow \begin{aligned} 4x+y &= 5x \\ x+4y &= 5y \end{aligned}$$

$$\Rightarrow \begin{aligned} -x+y &= 0 \\ x-y &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} x-y &= 0 \\ x-y &= 0 \end{aligned}$$

$$\therefore y=0$$

In echelon form, there is only one equation for two unknowns, therefore the system has a free variable y .

$$\text{Let } y = 1 \Rightarrow x = 1$$

$\therefore$ Eigenvector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for $\lambda = 5$

Let P be the matrix of eigenvalues of A .

$$\therefore P = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$|\det P| = -1 - 1 = -2$$

$$C_{11} = (-1)^{1+1} (1) = 1$$

$$C_{12} = (-1)^{1+2} (1) = -1$$

$$C_{21} = (-1)^{2+1} (1) = -1$$

$$C_{22} = (-1)^{2+2} (-1) = -1$$

$$\text{Adj}(P) = (P^c)^T = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}^T = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

$$\therefore P^{-1} = \frac{\text{Adj } P}{\det P}$$

$$= \frac{1}{-2} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

Then A can be written as an eigen decomposition

$$A = PDP^{-1}$$

$$\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$= -\frac{1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3+0 & -3+0 \\ -5 & -5 \end{pmatrix}$$

$$= -\frac{1}{2} \begin{pmatrix} -3-5 & 3-5 \\ 3-5 & -3-5 \end{pmatrix}$$

$$= -\frac{1}{2} \begin{pmatrix} -8 & -2 \\ -2 & -8 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} = A \text{ (proved)}$$

$$\therefore A = PDP^{-1} \text{ (proved)}$$

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